



The podocyte slit-diaphragm: target of anti-nephrin antibodies

Laurence H. Beck and David J. Salant

Purpose of review

This review is prompted by substantial new information on the composition and structure of the podocyte slit-diaphragm and the identification of anti-nephrin antibodies in patients with acquired diseases of the glomerular podocyte, including steroid-sensitive nephrotic syndrome of childhood, minimal change disease and some cases of primary focal and segmental glomerulosclerosis (FSGS).

Recent findings

New methodologies including multi-epitope affinity purification, high resolution proteomics and cryo-electron tomography reveal that the slit-diaphragm consists of a multi-layered fishnet-like structure made up by nephrin and neph1 homodimers as well as several co-assembled proteins with signaling properties. Insights from clinical studies and experimental models support the concept that pathogenic antibodies engaging the extracellular domain of nephrin disrupt the integrity of the slit-diaphragm and signal changes to the podocyte cytoskeleton that lead to foot process effacement and proteinuria.

Summary

These findings raise several questions to be answered experimentally, including the nephrin epitopes targeted by the antibodies and the way in which they alter the slit-diaphragm and podocyte architecture and induce nephrin endocytosis. They also highlight the need for a reliable and widely available assay for anti-nephrin antibodies, which would have diagnostic and therapeutic impact on diagnosis, prognosis and therapy.

Keywords

anti-nephrin antibodies, focal and segmental glomerulosclerosis, minimal change disease, podocyte, primary and recurrent, slit-diaphragm

INTRODUCTION

Given the recent focus on anti-nephrin antibodies in patients with acquired podocytopathies, especially steroid-sensitive nephrotic syndrome of childhood, minimal change disease (MCD) and some cases of primary FSGS, and the fact that nephrin is a key constituent of the podocyte slit-diaphragm (SD), we reasoned that a review of the SD itself is timely. This is especially true because substantial new information on the composition and structure of the diaphragm that bridges filtration slits between adjacent podocyte foot processes demonstrate that it is more than a simple structural barrier to macromolecule permeation. Signal transduction to the podocyte cytoskeleton by nephrin itself or its interacting partners likely influences the buttressing structure of the foot processes, which in turn exert compressive force on the glomerular basement membrane (GBM) and maintain its selective permeability [1]. Focal or diffuse effacement of podocyte foot processes with loss of their buttressing force is a universal morphological

finding in proteinuric glomerular diseases and may be the end result of signals transduced from the SD.

Cells destined to become podocytes first appear in the S-shaped body during glomerular development as simple, cuboidal epithelial cells with apical tight junctions linked to zonula occludens-1 (ZO-1) on their cytoplasmic surface [2,3]. As the cells mature, the tight junctions migrate basally and eventually disappear to be replaced by the SDs that span the filtration slits between the interdigitating foot

Renal Section, Department of Medicine, Boston Medical Center and Boston University Chobanian and Avedisian School of Medicine, Boston, Massachusetts, USA

Correspondence to David J. Salant, MD, Department of Medicine Boston University Chobanian and Avedisian School of Medicine Evans Biomedical Research Center, 804, 650 Albany Street Boston, MA 02118, USA. Tel: +1 617 638 7336; e-mail: djsalant@bu.edu

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KEY POINTS

- The podocyte slit-diaphragm (SD) is essential to the integrity of the selective glomerular filtration barrier and the maintenance of normal podocyte foot-process architecture.
- New cryo-electron tomography studies reveal that the SD is made up of a fishnet-like structure of crisscrossing nephrin and neph1 heterodimers that provides flexibility and porosity, in addition to components covering several functions predominantly involved in signaling and formation of cellular junctions.
- That nephrin may serve as a target of pathogenic antibodies is evident from the occurrence of post-transplant nephrotic syndrome in children with congenital nephrotic syndrome from nephrin deficiency, and from the transfer of anti-nephrin antibodies to experimental animals.
- Mechanistic studies demonstrate that antibodies that target the extracellular domain of nephrin signal changes to the podocyte cytoskeleton that lead to proteinuria.
- Similar mechanisms may account for the development of proteinuria induced by anti-nephrin antibodies in patients with podocytopathies such as minimal change disease and primary focal and segmental glomerulosclerosis.

processes of adjacent podocytes that retain ZO-1 at the site of SD attachment.

STRUCTURE AND COMPOSITION OF THE SLIT-DIAPHRAGM

The structure and composition of the SD have been subject to investigation since it was first identified by transmission electron microscopy. In 1955, Yamada described the mouse SD as “a delicate gossamer-like membrane about 30 Å (3 nm) thick” [4]. In 1974, based on transmission electron micrographs of freeze-fractured tissue, Rodewald and Karnovsky described it as a continuous zipper-like structure [5]. They determined that the dimensions and spacing of the cross bridges defined a uniform population of cylindrical channels 70–84 Å (7.0–8.4 nm) in diameter. These findings led to the concept of the SD as an isoporous structure that serves as the final barrier to the permeation of macromolecules from plasma into the urinary space [6].

The zipper-like model proposed by Rodewald appeared to hold up to scrutiny with the discovery of nephrin as an integral component of the SD [7]. In fact, it was considered that the zipper-like structure

with intervening pores was created by the trans interaction of molecules of nephrin and the shorter homologues of nephrin, neph1 and neph2, from adjacent foot processes across the filtration slits, a model that appeared to be confirmed by electron tomography. However, subsequent studies showed the SD to be a more complex structure than a simple nephrin bridge, composed of several other constituents, including P-cadherin, FAT1 as well as anchoring proteins like podocin and CD2-associated protein (CD2-AP), signaling molecules and links to the actin cytoskeleton. Notably, earlier studies that used fresh-frozen unfixed tissue and a deep-etching replica method had challenged the zipper-like structure as a tissue fixation artifact [8]. Which brings us to the era of multi-epitope affinity purification (meAP), high resolution proteomics and cryo-electron tomography.

The interacting proteins that make up the SD were most recently defined in a study of mouse and rat glomerular membrane fractions using a meAP approach with antibodies targeting nephrin, neph1, and podocin [9]. The study revealed a co-assembled high-molecular-weight protein network comprising the three target proteins as well as several previously known and unknown transmembrane and cytoplasmic SD components covering several primary functions predominantly involved in signaling and formation of cellular junctions. The analysis identified a collection of about 25 additional proteins, mostly membrane-associated or transmembrane proteins of different classes that co-assemble with the core components and form a network-like arrangement of high molecular mass. Selective, post-natal induced knockout experiments of nephrin, neph1 and podocin in mice revealed differences in both the dynamics of albumin permeability and variations in the effects on different components of the protein interactome. Additionally, the functional significance of several of the newly identified proteins, including constituents with enzymatic activity, receptor-triggered signaling and scaffolding function, was established by knock-down experiments in *Drosophila* nephrocytes. The clinical veracity of this exercise in protein chemistry is evident from the identification of several interacting proteins known to underlie hereditary nephrotic syndrome, including dendrin, PTPRO and MAGI2 [10–12] as well as the tyrosine kinase Fyn and transmembrane receptor ROBO2, two established nephrin/neph1 interactors [13,14]. However, other proteins such as CD2-AP and the TRPC6 ion channel, that are known to be the cause of hereditary nephrotic syndrome or to interact with the core SD complex (such as the adaptor protein NCK1, JAM-A (junctional adhesion molecule A) and ZO-1) [15] were not captured by the meAP

approach. This opens the possibility that there may be other SD constituents yet to be identified.

THE STRUCTURE OF THE SLIT-DIAPHRAGM RESEMBLES A BILAYERED FISHNET

Rather than the zipper-like structure previously described, more recent morphological studies using nephrin and neph1 knockout mice have documented that the SD is a flexible, multilayered protein scaffold [16]. In addition, detailed cryo-electron imaging studies of mouse podocyte nephrin and

neph1 [17^{***}] and their *Drosophila* nephrocyte orthologs Sns and Kirre [18^{***}] revealed criss-crossing strands of the proteins resembling a fishnet-like structure of the SD (Figs. 1 and 2). The calculated lateral dimensions of the spaces in the fishnet of ~12.3 nm are similar as the purported “pores” identified in scanning EM images (12.1 nm) [19] but moderately larger than those in the zipper model (7.0–8.4 nm) [5]. This fishnet-like arrangement is thought to provide both stability and flexibility essential for glomerular filtration. Moreover, silencing Sns (nephrin ortholog) expression in *Drosophila*

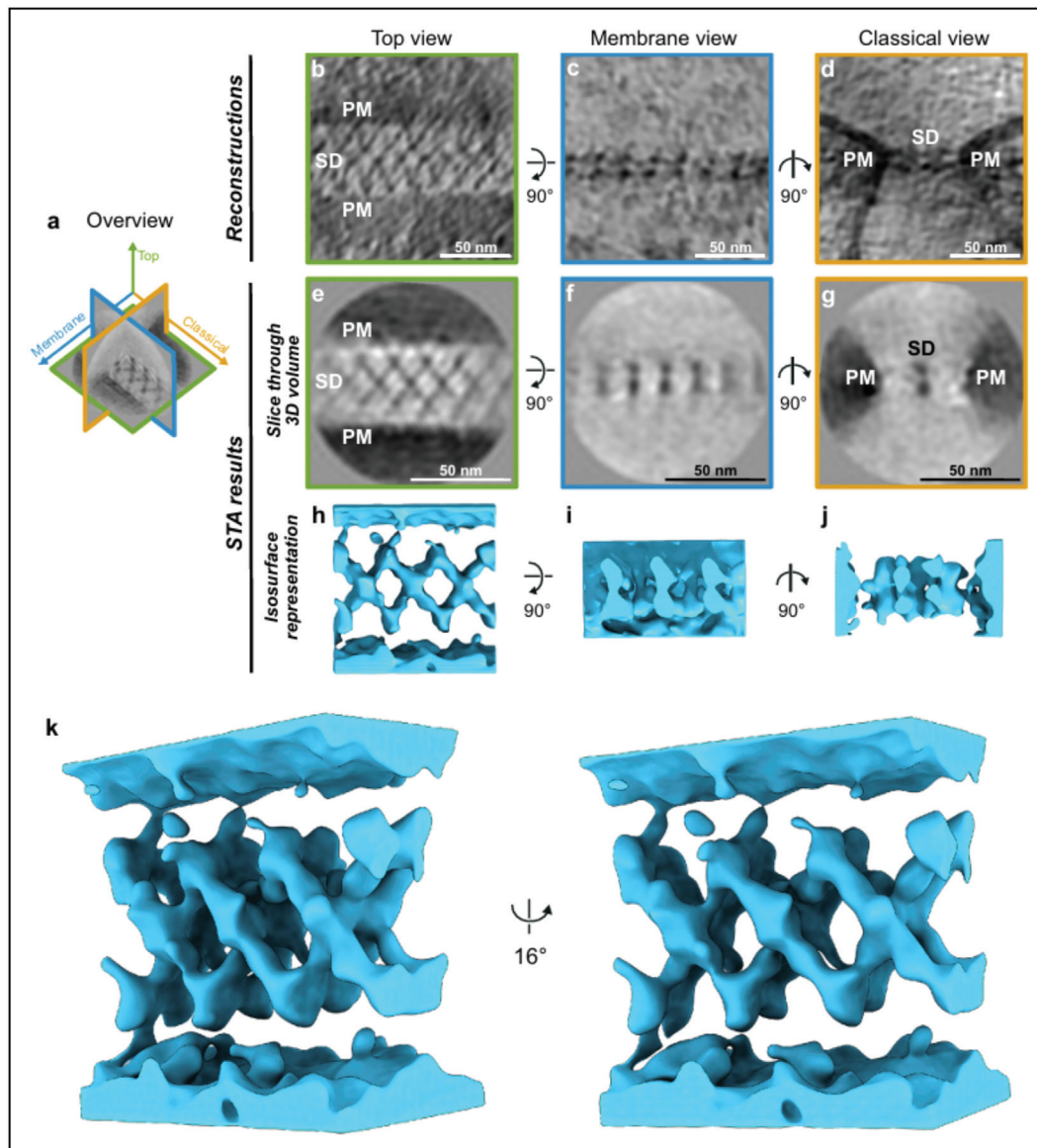


Figure 1. Cryo-ET views of the *Drosophila* nephrocyte slit diaphragm (SD) reveal a bilayered fishnet-like structure. Tomographic reconstructions from three perspectives are shown (b–g). When seen from above the individual strands are seen to crisscross the space between the two plasma membranes creating the fishnet pattern. From the side (d and g; classical view), two parallel electron-dense lines, reflecting two layers of the SD, are visualized. When viewed from an oblique angle (k), the bilayered architecture of the SD is revealed with each of the layers resembling a fishnet. From Moser *et al.* [18^{***}].

caused ectopic dislocation of the fishnet pattern, which may represent the alteration that occurs in human congenital nephrotic syndrome in which nephrin is absent. The fishnet models developed by Birtasu and Moser are based on the >100-fold abundance of nephrin and neph1 as compared to other SD constituents extracted from rodent glomeruli, as well as the electrostatic interaction of their complementary differentially charged Ig domains. However, it remains to be determined how other SD components, like those identified by Kocylowski *et al.* [9] and those whose absence causes hereditary nephrotic syndrome, are integrated into this fishnet-like structure.

THE SLIT-DIAPHRAGM AS A TARGET OF INJURY

So, if nephrin is an essential constituent of the SD, which, in turn, is critical for the maintenance of glomerular permselectivity, is it a target of anti-nephrin antibodies in hereditary or acquired proteinuric kidney diseases? And if so, how does it happen?

NPHS1, the gene encoding nephrin, was first discovered in 1998 among neonates in Finland with autosomal recessive congenital nephrotic syndrome (CNS) [20]. Two *NPHS1* variants account for >90% of the Finnish cases but several other variants have been identified in non-Finnish families with CNS. Most are unique to each family and comprise missense, nonsense, and splice site mutations as well as small deletions and insertions [21]. Fin-major, the more common variant that accounts for 78% of patients with CNS in Finland, is a frame shift deletion of two base pairs in exon 2 that generates an early stop codon and encodes a protein of only 90 amino acids. As a result, there is no expression of the mature protein in glomeruli.

Infants with CNS and the homozygous Fin-major genotype are of particular interest in the context of recent studies on the pathogenic potential of anti-nephrin antibodies, especially after kidney transplantation. This is because they may develop post-transplant nephrotic syndrome, which in some cases was shown to be associated with anti-nephrin antibodies, presumably due to alloantibodies to the novel, full-length nephrin molecule in the allograft [22,23]. In a 2014 review of published cases, Holmberg and Jalanko concluded that recurrence occurs in about 30% of Fin-major/Fin-major patients of which 70% have measurable anti-nephrin antibodies [24]. Moreover, plasma exchange combined with cyclophosphamide and anti-CD20 antibodies was effective therapy for such cases. While the Fin-major mutation is rare outside Finland, these observations provide potential proof of concept that circulating anti-nephrin antibodies may be pathogenic.

More direct evidence that anti-nephrin antibodies are able to cause proteinuria derives from experimental studies in rats injected with mAb 5-1-6, a murine monoclonal antibody that binds to the SD where it engages the extracellular domain of nephrin [25-27]. Immediately after mAb 5-1-6 injection, rats develop severe proteinuria as the murine immunoglobulin G (IgG) binds nephrin which is then partly redistributed into cytoplasmic vesicles accompanied by a decline in SD nephrin immunoreactivity [25,28]. Simultaneously, ZO-1 is lost from the podocyte. Within days, nephrin and ZO-1 immunoreactivity of the podocytes recover as mAb 5-1-6 is shed or degraded and proteinuria subsides. Mechanisms underlying these effects of the anti-nephrin mAb were recently identified by additional studies from the Niigata investigators [29^{***}]. These disclose a complex series of signals transduced from nephrin to the podocyte upon mAb 5-1-6 (hereafter anti-nephrin mAb) engagement that include phosphorylation of nephrin and its SD partner ephrin-B1 [30] in a TRPC6-mediated Ca^{2+} influx-dependent manner, leading ephrin-B1 to dissociate from nephrin and also from its intracellular binding partner Par6, a protein instrumental in controlling cell polarity [29^{***}]. Release of Par6 induced the activation of cdc42, a small Rho family GTPase and regulator of cell polarity, and actin cytoskeleton dynamics. These events might explain changes in podocyte morphology and alteration in barrier function. Interestingly elevated cdc42 activity also activated Stat3 leading to the expression of the tight junction molecule claudin1, which might correspond to the known reversion of filtration slits to tight junctions in some proteinuric diseases [30]. The authors also note that like MCD, the anti-nephrin monoclonal antibody (mAb) could not be detected by standard immunofluorescence when abnormal proteinuria appeared 5 days after rats were injected with the minimum nephritogenic dose of anti-nephrin mAb dose.

The cellular events triggered by anti-nephrin antibodies described by Zhang *et al.* differ from those described by Verma *et al.* and Jones *et al.* as resulting from nephrin ectodomain engagement, which activated the Src family tyrosine-kinase Fyn and resulted in nephrin and neph1 phosphorylation, Nck recruitment and actin polymerization. Moreover, selective deletion of Nck from podocytes resulted in defective foot processes formation and proteinuria in transgenic mice [31]. Although the two mechanisms of cell signaling activated by anti-nephrin mAb and ectodomain engagement are quite distinct, they are not mutually exclusive. Nephrin engagement with its SD partners may be essential for maintenance of the actin cytoskeleton and podocyte architecture while anti-nephrin activates a more disruptive signaling

cascade leading to the effacement of foot processes and the loss of their buttressing effect.

Findings similar to those with anti-nephrin mAb were described in mice immunized with nephrin to generate a model of anti-nephrin associated podocytopathy [32]. The mice developed proteinuria and histologic findings resembling MCD. Proteomic analysis revealed nephrin phosphorylation “at a site known to be involved in actin assembly”. The authors proposed that this was a signal from the SD that induced foot process effacement.

Thus, substantial evidence from human genetic nephropathy and experimental models indicates that anti-nephrin antibodies can cause podocyte injury and proteinuria by altering the integrity of the SD and glomerular filtration barrier. The discovery of human anti-nephrin autoantibodies in cases of MCD, primary FSGS, and recurrent FSGS in the kidney allograft suggests that this is not limited to the unique circumstances described above but may rather be a common pathogenetic mechanism with potential diagnostic and therapeutic implications.

ANTI-NEPHRIN ANTIBODIES IN ACQUIRED HUMAN PODOCYTOPATHIES

The landmark articles describing anti-nephrin antibodies in both childhood and adult MCD and primary FSGS [32,33] have been well summarized elsewhere [34[■],35[■],36] but initial observations from these papers are relevant to the discussion above. Prior to finding evidence of circulating autoantibodies, the renal pathology group in Boston led by Helmut Rennke had consistently observed a “delicate punctate staining for IgG” over the podocyte cell body in a subset of patients with MCD [33]. Super-resolution structured illumination microscopy (SR-SIM) revealed two distinct patterns in these kidney biopsy specimens: IgG in close apposition to the GBM in a fine punctate or curvilinear pattern (likely tracing the actual slit diaphragm) or in more apically located punctate or vesicular clusters [33]. These latter structures co-localized nearly perfectly with nephrin but not with three other podocyte proteins. Based on the entirety of their data, the authors speculated that these punctate structures represented the in situ binding of anti-nephrin, potentially able to disrupt the integrity of the SD and cause redistribution of nephrin and its bound autoantibody. This redistribution could be indicative of nephrin endocytosis as previously described in the literature or the recently described process of auto-immunoglobulin-triggered extracellular vesicle formation [37[■]] since these particles were found in the urine of MCD patients and contained both nephrin and IgG.

Another relevant point is that the IgG in the punctate GBM and podocyte staining in these MCD cases was predominantly IgG1 and IgG2 (which are bivalent) [33], and not the IgG4 (functionally monovalent) subclass found in other non-inflammatory autoantibody mediated diseases such as membranous nephropathy. If one assumes that this IgG is indeed anti-nephrin, such a finding would indicate that a single autoantibody could cross-link two nephrin molecules, possibly initiating a signaling event. The rat model with anti-nephrin mAb required infusion of bivalent anti-nephrin to induce proteinuria in a complement-independent manner, as a single Fab arm was not able to induce proteinuria [38].

Detection of circulating anti-nephrin has not been standardized in the research setting and is not yet available for clinical use. Typical methods useful for the detection of other human autoantibodies such as immunoblotting with patient serum, standard ELISA, or cell-based techniques have not been successful [39]. Instead, Hengel *et al.* were able to achieve high sensitivity for the detection of circulating anti-nephrin in MCD (especially if nephrotic and untreated with immunosuppression) as well as in 9% of primary FSGS cases, using an enhanced detection method [32]. This assay was a hybrid method using patient serum to first immunoprecipitate recombinant human nephrin and then quantitate the tagged nephrin molecule by ELISA. It is not known if these enhanced techniques are required because of low titers of anti-nephrin, weak affinity of the paratope-epitope interaction, or other non-traditional mechanisms of nephrin-IgG binding. Other issues such as the species of the cell-line (human vs mouse) used to produce the recombinant nephrin appear to be important, since specific glycan moieties introduced into the antigen by non-human species can cause non-specific binding by naturally-occurring anti-glycan human antibodies [39,40].

While anti-nephrin antibodies may be the dominant anti-slit diaphragm antibodies in MCD and primary FSGS, others have also demonstrated evidence of antibodies to Kirrel1 (nepH1) and podocin [41] and to nepH3 [40], occasionally in combination with anti-nephrin. As these other molecules interact with nephrin in the SD, autoantibodies directed against them may synergize with anti-nephrin to cause rapid-onset proteinuria or induce a similar response on their own. Podocin in its normal state, however, does not present extracellular epitopes that could react with circulating antibodies, but a genetic variant (P118L) does project its C-terminus into the extracellular space, so it is possible that such antibodies could have a biologic effect in certain situations.

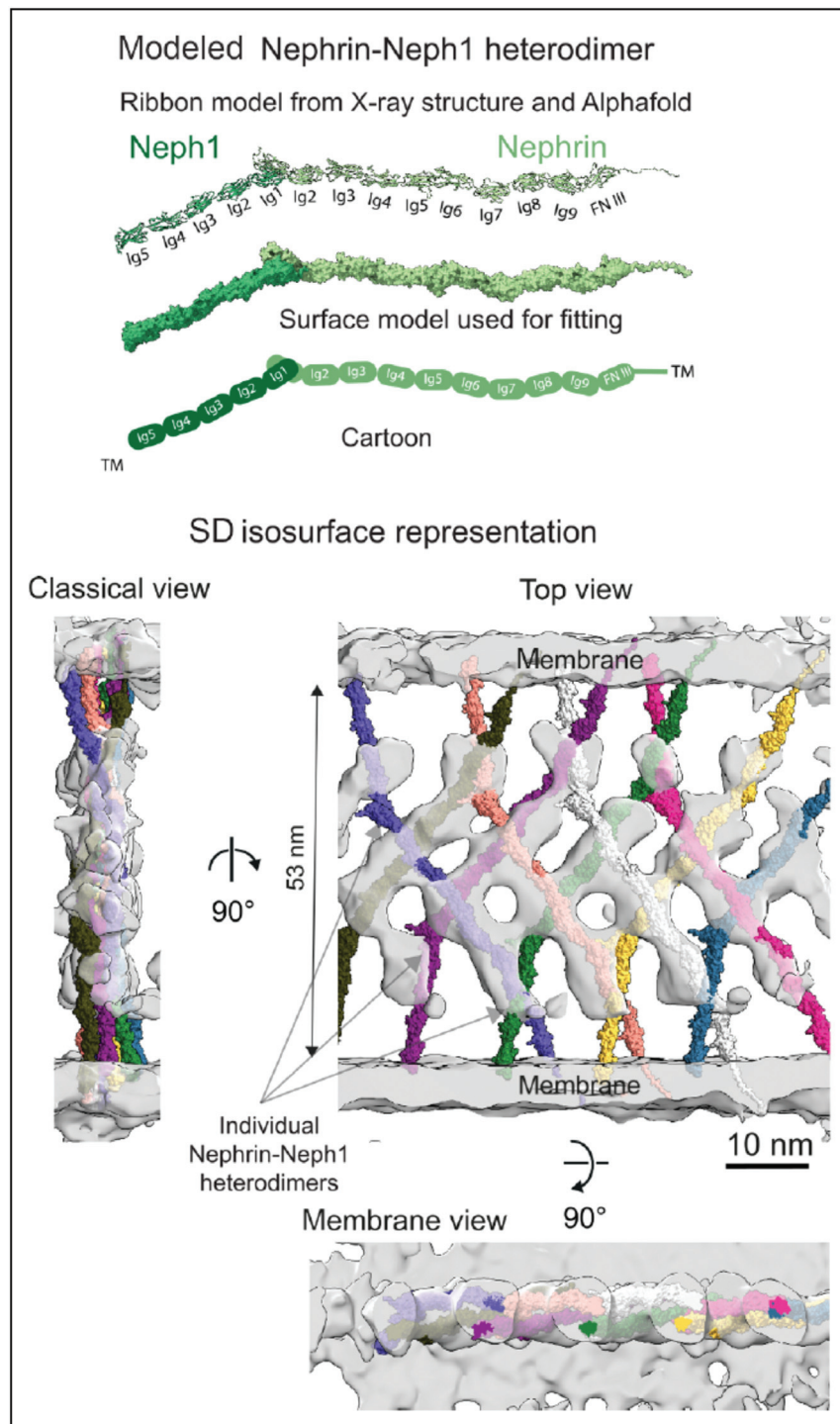


Figure 2. Based on cryo-ET, the mouse slit diaphragm (SD) is shown to be a network of crisscrossing nephrin-Neph1 heterodimers that create a fishnet pattern. Upper panel: the extracellular parts of the nephrin (light shade) and neph1 (dark shade) heterodimer as predicted from AlphaFold are shown as a ribbon and surface model. The binding of the Ig1 domains of the two proteins to form a heterodimer is modeled on the crystal structure of the *C. elegans* SYG-2–SYG-1 interface. A cartoon of the heterodimer shows the individual Ig domains and the transmembrane region (TM). Lower panel: Nine nephrin–Neph1 heterodimers (each shown in a different color to maximize contrast) are fitted to the tomogram average (shown in transparent shade). The spacing between the heterodimers is 12 nm from center to center, leaving about $7 \times 7 \text{ nm}^2$ of open space. Each heterodimer spans the distance between the plasma membranes of adjacent podocytes. A fishnet pattern is created by individual heterodimers crisscrossing each other. From Birtasu *et al.*[17[■]].

There have been whisperings that the presence of anti-nephrin during clinically active disease may merely be an epiphenomenon, occurring in response to podocyte damage but not actually causing it. Some have cited a lack of clear temporal precedence (anti-nephrin prior to proteinuria) and the very rapid remissions (both clinical and immunological) that occur “too soon” and defy the normal half-life of IgG, especially in the face of resolving selective proteinuria [42]. The field is accustomed to other autoantibodies such as anti-PLA2R declining and disappearing over the course of weeks to months. But while a gradual build-up of immune complexes and complement activation is required prior to clinical proteinuria in MN, it is biologically plausible that low levels of anti-nephrin could rapidly induce proteinuria through direct effects on nephrin and the SD, which would lead to a more immediate correlation of antibody and proteinuria. It has been shown that transplantation of a kidney allograft into patients with circulating anti-nephrin can cause immediate histopathological changes one hour after transplantation which were not present in the pre-perfusion biopsy, including punctate co-localization of IgG and nephrin and phosphorylation changes in nephrin [43].

Passive transfer of disease to experimental animals is a gold standard for establishing pathogenicity of autoantibodies. The only published passive transfer model for human anti-nephrin describes the administration to a rabbit of total IgG obtained from plasmapheresis fluid from a 71-year-old woman with MCD, taken at a time of peak circulating anti-nephrin antibodies [44]. When compared to the rabbit treated with control IgG, the experimental rabbit exhibited trace albuminuria at 3 days post-injection that was maximal at day 5. There was also partial effacement of podocyte foot processes by electron-microscopy (EM) at day 5, and discontinuous punctate staining for human IgG along the GBM, with occasional cytoplasmic punctate staining. IgG eluted from day 5 glomeruli could detect rabbit nephrin. While suggestive, the results do not exclude other anti-slit antibodies (possibly anti-podocin or anti-neph3 as described above) that may have also been present with anti-nephrin) and affinity depletion of anti-nephrin from the IgG was not performed. The delayed (5 days) effect of the administered anti-nephrin-containing IgG and why there was still prominent GBM staining along all visualized GBM at 5 days may reflect differential action of these antibodies against rabbit nephrin.

CONCLUSION

How do we integrate the presence of anti-nephrin antibodies and rapid-onset proteinuria in MCD and primary FSGS with the newly proposed fishnet

structure of the SD? The multiple interaction points between nephrin and neph1 are proposed to allow individual molecules to be removed (probably by endocytosis) without impacting overall integrity of the SD [17]. However, as has been shown in the rat with anti-nephrin mAb [29] (and references therein), low levels of circulating anti-nephrin (despite targeting only a minority of molecules within the larger structure) may be sufficient to disrupt molecular interactions, cross-link molecules by their bivalent nature and trigger endocytosis or signaling, leading to a coordinated collapse of the SD structure. There is some evidence that human anti-nephrin targets the first Ig domain of nephrin [41] and thus could potentially disrupt the nephrin-Neph1 nodes tethering the two sides of the slit pore. Many exciting questions remain that can be addressed experimentally. Where, in fact, are the epitopes targeted by anti-nephrin antibodies? Can we affinity purify or clone human anti-nephrin antibodies to better define binding sites, demonstrate an ability to disrupt nephrin homo- or heterodimers in vitro, and induce nephrin signaling and endocytosis in model systems? Apart from these exciting opportunities to define pathomechanisms of MCD and primary FSGS, the field is in critical need of an optimal assay system for clinical testing for anti-nephrin antibodies which would have impact on diagnosis, prognosis, choice of therapy and decisions about kidney transplantation and recurrent disease.

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Conflicts of interest

There are no conflicts of interest.

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